Operating Systems

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Agenda

- ✔ Foundations of Operating Systems
- ✔ Windows 10
- ✔ Unix/Linux
- ✔ Mac OS X
- ✔ Android OS
- ✔ Apple iOS
- ✔ Golden Images

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Foundations of Operating Systems

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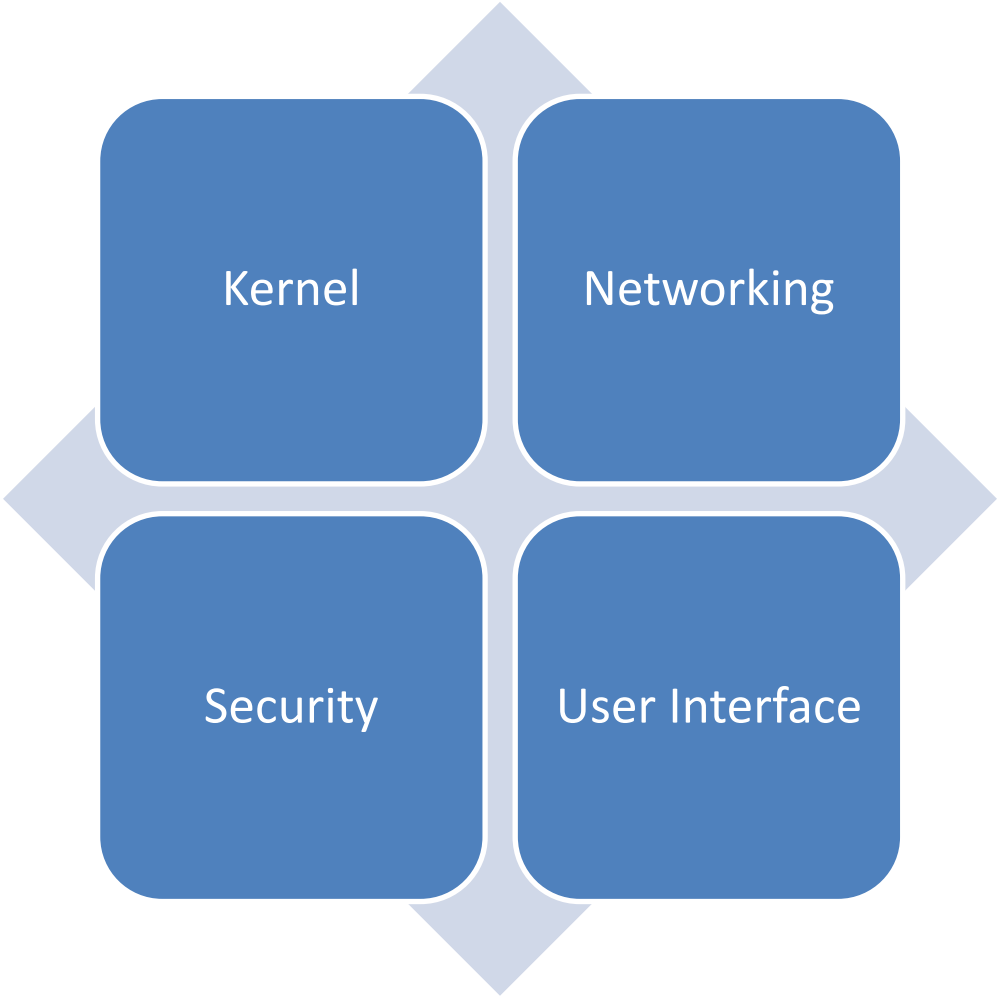
Components



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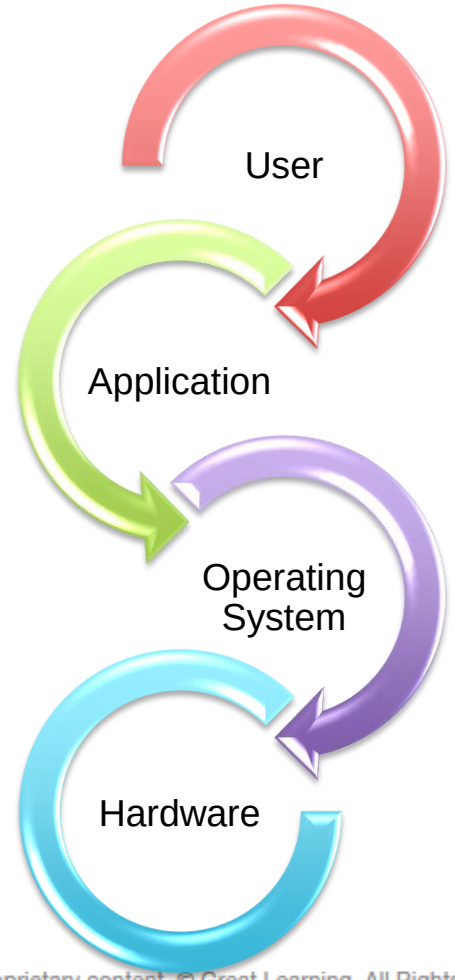
Operating System

Intermediary between programs and computer hardware



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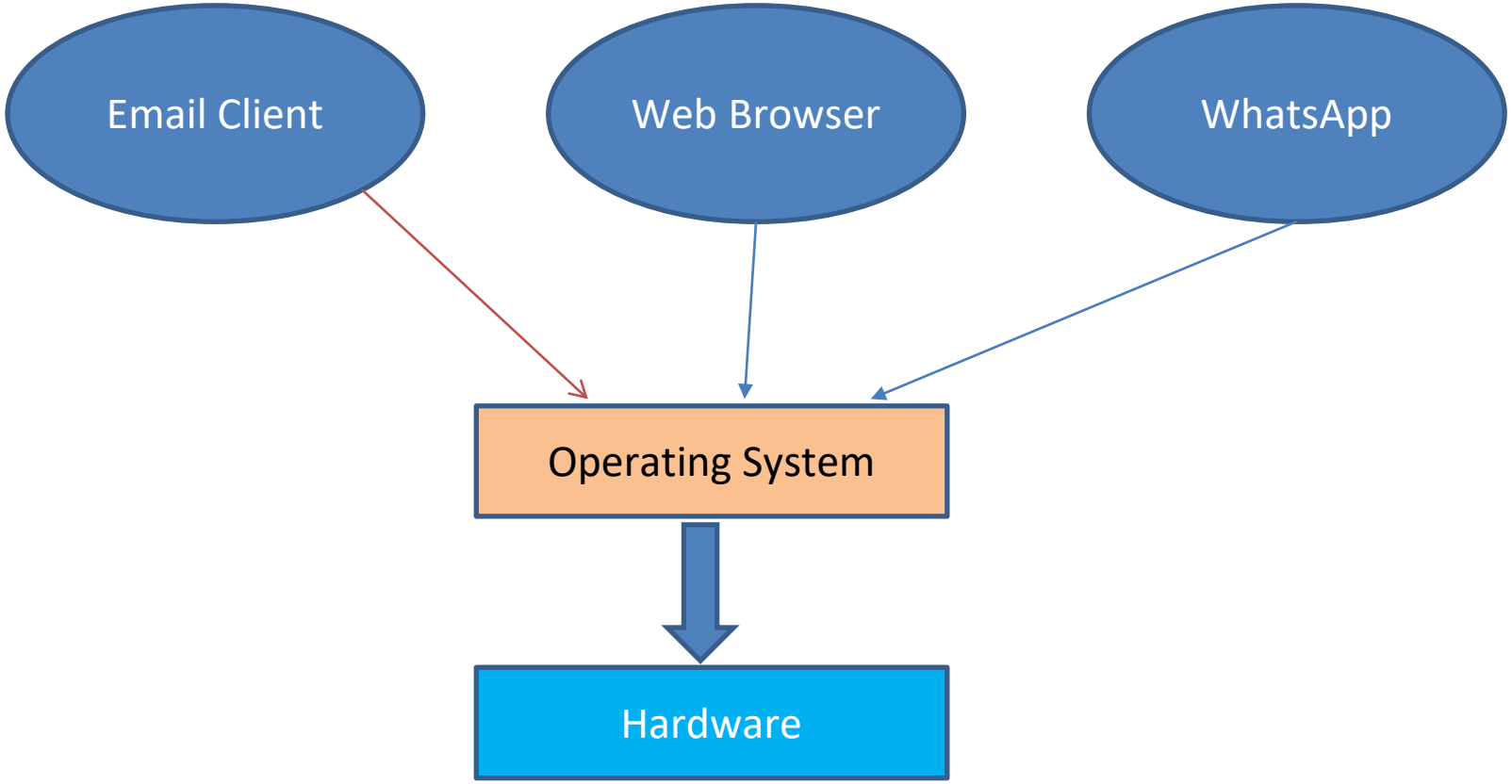
Operating System



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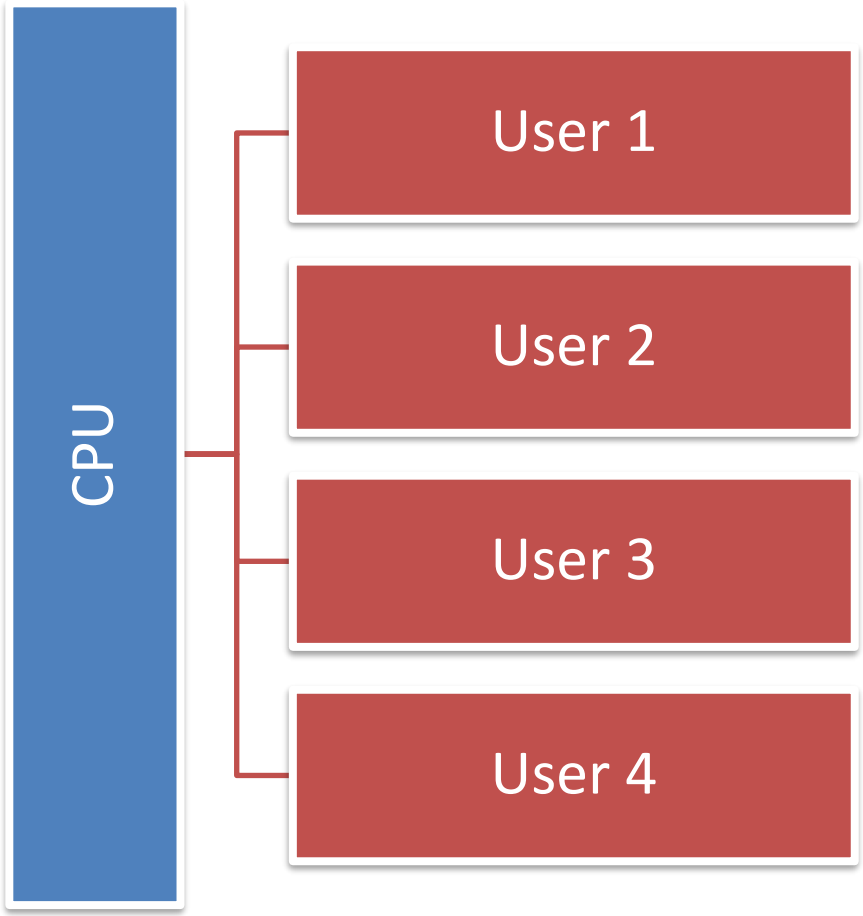
Time sharing concept



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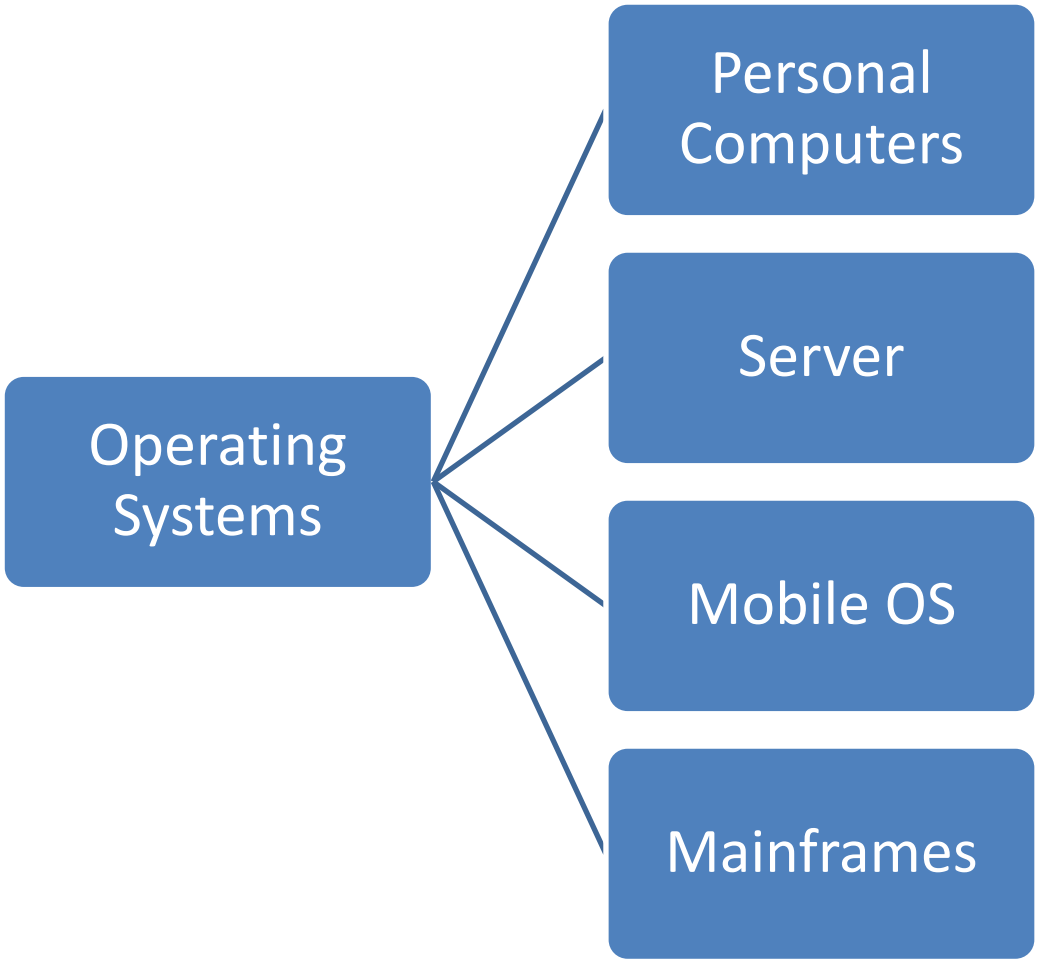
Operating System flavors and types



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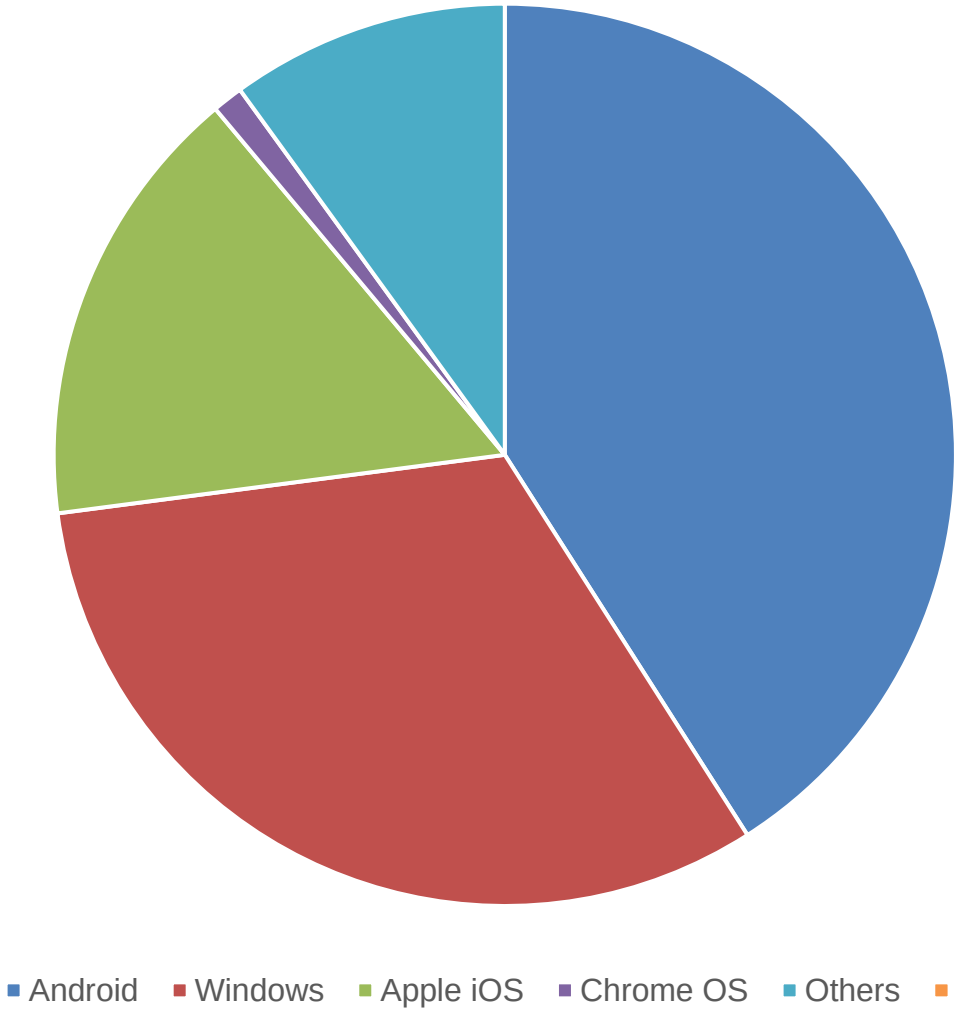
Different segments of Operating Systems

- Personal computer
- Servers
- Mobile OS



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Windows 10

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Features that are key to security

- File and Folder Structure
- Control Panel
- Services Management
- Windows Updates
- Event Management
- Task Management

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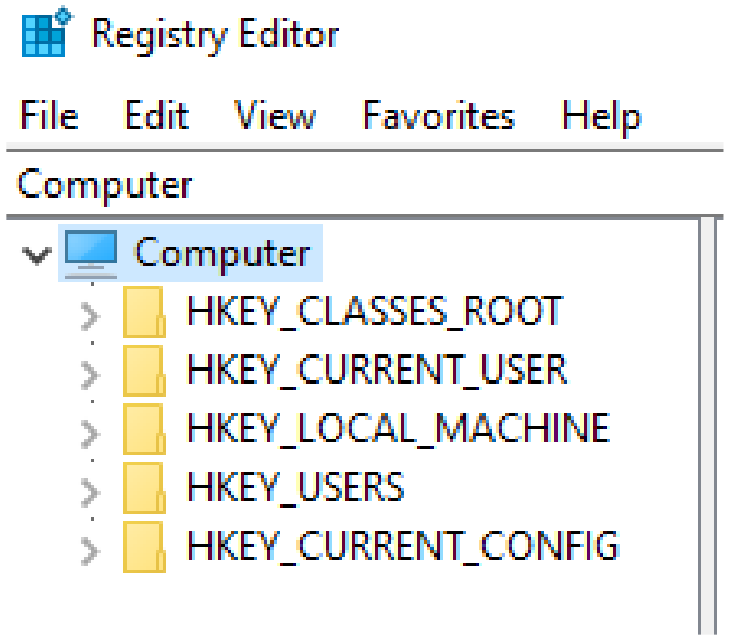
Registry

- Stores configuration settings in Binary format
- Must edit with special programs
- Hierarchical structure
- Allows multiple users a simultaneous access
- Allows Per user settings



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Registry

- Divided into separate sections or “Hives”
  - HKEY_CLASSES_ROOT
    - Registered applications and file associations
  - HKEY_CURRENT_USER
    - Settings for the current user of the system
  - HKEY_LOCAL_MACHINE
    - Local machine settings
  - HKEY_USERS
    - Settings specific to each user of the system
  - HKEY_CURRENT_CONFIG
    - Configuration data for current hardware profile

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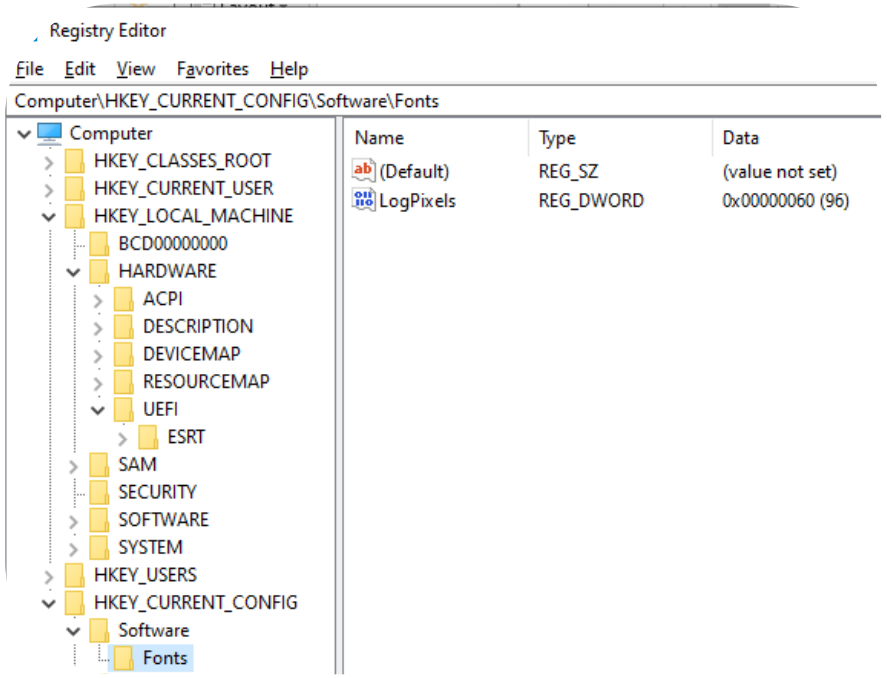
Common registry value types

- REG_SZ – String (Text)
- REG_MULTI_SZ – Multiple Strings
- REG_DWORD – A number between 0 and 4,294,967,295
- REG_BINARY – Binary data



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Windows Server 2016

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Windows Server 2016 editions

- Windows Server 2016 platforms:
  - Windows Server 2016 Essentials Edition
  - Windows Server 2016 Standard Edition
  - Windows Server 2016 Datacenter Edition
- Additional Windows Server 2016 platforms targeted for specific-purpose niches:
  - Windows Server 2016 Multipoint Premium Server
  - Windows Storage Server 2016
  - Windows Hyper-V Server 2016

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Features

- Server Manager
- Security
- Clustering
- Enhanced Web services
- Windows Server Core and Nano Server

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Features

- Windows PowerShell
- Virtualization
- Reliability
- Multitasking and multithreading
- Containers

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MAC OS X

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MAC OS X

- Primary OS for Apple Computers
- Based on Unix
- Offers a simple user interface
- macOS Montery is the latest version (Mac OS X V 12)

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Features

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Features

- Spotlight search
- Mission control
- Keychain
- Dock
- System preferences
- AppStore
- Finder
- Applications

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Android

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Android OS

- It is an open-source system.
- Design is based on Linux Kernel and other open source software.
- Used for mobile devices, for instance tablets, smartphones and mobile devices with touchscreen.

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Layers in Android OS

greatlearning
Power Ahead

Application Framework

- Programmed In Java
- Performs as per Principle of Least Privilege

Binder IPC

- Enables communication
- Implementation is via Android OS Kernel

Android Services

- Types- Started and Bound Services
- Eg: Camera, Media Player, Search, Activity Manger Services

Hardware Abstraction Layer

- Enables insertion of functionality
- Camera, Graphics, Audio HALs

Linux Kernel

- Advanced version of Linux for additional features
- Ensures Open Source system

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Features

- Customizable widgets
- Automation
- Applications
- Near field communication (NFC)
- Customizable Read-Only memory
- Infrared communication
- Alternate keyboards
- Voice based features

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Apple OS X

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About OS X

- macOS Monterey is the latest version
- Key Features:
  - Time machine
  - Migration assistant
  - Utilities
  - Spotlight
  - Keychain
  - Activity monitor

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Golden images

A golden image provides the template which a virtual machine (for example, AWS EC2 instances) is created from. It may also be referred to as a base image or an image template. Think of it as a snapshot copy of an operating system that can be launched as a new virtual machine.

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Golden images

- Usually, a golden image will contain
  - A Windows or Linux operating system installation
  - The latest security patches and updates
  - Configuration specific to your environment
  - Software specific to your environment
  - Security hardening settings, if required by your environment

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Summary

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